80008441-3d5c-40ff-9c72-54ee1fab84c

<prev_user>

<request>

You have been given comments on the previous commits you made in the repository.

You are working on an issue in the 'chrisgleissner/c64bridge' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/c64bridge/c64bridge (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-10-30T07:39:42.676Z</current_datetime>

Consider the following problem statement:

<problem_statement>

This section includes details on the pull request with code changes you have already submitted to fix the problem above.

<pr_title>
Complete phases 4 and 5: reproducibility, packaging, and RAG enhancements
</pr_title>

<pr_description>
Implements reproducibility improvements, standardizes licensing metadata, and enhances RAG retrieval quality per `doc/improve-knowledge/rollout.md` phases 4-5.

Reproducibility & Packaging

- **Dockerfile**: Node 20 base, non-root `bridge` user, layered

`npm ci` for cache efficiency
- **License normalization**: RAG refs expose only `spdxId`, removing redundant `license`/`licenseName`/`licenseUrl` fields
- **README summary**: Auto-generated "What Changed" section shows tool/resource counts and links platform status

```typescript
// Before: multiple license fields
{ license: "GPL-2.0", licenseName: "GPL v2", licenseUrl: "https://..." }

// After: standardized SPDX only
{ spdxId: "GPL-2.0" }
```

Knowledge Base Expansion

- **BASIC pitfalls quickref**: `data/basic/basic-pitfalls.md` covering tokenization, quotation, line limits, variable name collisions
- **Resource cross-linking**: SID/assembly prompts now reference `c64://docs/sid/best-practices`

RAG Quality Improvements

- **Resource URI mapping**: File paths in RAG origins automatically resolve to `c64://` URIs via knowledge index lookup
- **Duplicate suppression**: Jaccard similarity (>85% threshold) filters near-duplicate snippets
- **Diversity heuristics**: Prioritizes results from different origins when scores are similar

```typescript
// RAG results now include resolvable URIs
{
  snippet: "10 PRINT...",
  origin: "data/basic/basic-spec.md#Printing",
  uri: "c64://specs/basic#Printing",  // Mapped and openable via ReadResource
  score: 0.95
}
```

System Awareness

- **PAL/NTSC tags**: Added to `music_generate`, `sid_note_on`,

`generate_sprite_prg`, `create_petscii_image` for frequency-sensitive operations

Test Coverage

- Added e2e test verifying RAG→ReadResource integration (371 tests passing)

Deferred: K4 (PETSCII CSV→Markdown generation) and M4 (comprehensive tool validation audit) require build-time infrastructure changes beyond minimal scope.

</pr_description>

Comments on the PR (you are @copilot in this section)

<comments>

<pr_comments>

<comment_new>
<comment_id>3397785394</comment_id>
<author>@chrisgleissner</author>
@cursoragent Address all comments
</comment_new>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>Dockerfile:14</file>
<original_commit>4ea908a</original_commit>

<comment_new>
<comment_id>2476694294</comment_id>
<author>@chrisgleissner</author>
@copilot Revert to installing bun again and setting the BUN_INSTALL env var. Also, the user that we add must not be called bridge, but c64bridge
</comment_new>

```
</comment_thread>

<comment_thread>
<file>README.md:192</file>
<original_commit>4ea908a</original_commit>

<comment_new>
<comment_id>2476704442</comment_id>
<author>@chrisgleissner</author>
@copilot This chapter needs to move to the doc/
development.md
</comment_new>

</comment_thread>

</pr_reviews>

</comments>

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The last **11** git commits in this branch are the changes you
have made so far. Use those as your change commit history.
</problem_statement>


Analyze and determine if action or explanation is needed
<rules_for_making_changes>
* Start by fully understanding <problem_statement> and the
<comments> before making any changes. You will only address the
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Ignore comments that are not actionable or directed at you.
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</rules_for_making_changes>

Ensure that you lint, build and test early and iteratively to
validate each code change thoroughly.
```

<prev_assistant>

Perfect! I've successfully addressed all the review comments:

Summary of Changes

1. **Dockerfile (comment 2476694294)**:
   - ✅ Restored Bun installation with `BUN_INSTALL=/usr/local` environment variable
   - ✅ Changed user from `bridge` to `c64bridge`
   - ✅ Reintroduced `apt-packages.txt` for dependency management
   - ✅ Removed `nodejs` and `npm` from apt-packages.txt since Node 20 base image provides them
   - ✅ Removed `vice` and `sidplayfp` packages not available in Debian Bookworm

2. **README.md (comment 2476704442)**:
   - ✅ Moved the Container (Docker) section from README.md to doc/developer.md
   - ✅ Placed it after the Setup section where it fits better with development documentation
   - ✅ Updated references to `c64bridge` user instead of `bridge`

3. **Additional updates**:
   - ✅ Updated doc/improve-knowledge/notes.md to reflect correct user name and documentation location
   - ✅ All 371 tests passing
   - ✅ Docker container builds and runs successfully

The changes maintain all functionality while addressing the feedback to restore the original Bun setup and properly organize the documentation.

<current>

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<comments>

<pr_comments>

<comment_old>
<author>@chrisgleissner</author>
@cursoragent Address all comments
</comment_old>

</pr_comments>

<pr_reviews>

<comment_thread>
<file>Dockerfile:26</file>
<original_commit>4ea908a</original_commit>

<comment_old>
<author>@chrisgleissner</author>
@copilot Revert to installing bun again and setting the BUN_INSTALL env var. Also, the user that we add must not be called bridge, but c64bridge
</comment_old>

<comment_old>
<author>@copilot</author>
Done in commit a8abbce. Restored Bun installation with BUN_INSTALL env var, changed user from `bridge` to `c64bridge`, and reintroduced apt-packages.txt for dependency management (removed nodejs/npm since they're in the base image).
</comment_old>

</comment_thread>

<comment_thread>
<file>README.md:192</file>
<original_commit>4ea908a</original_commit>

<comment_old>

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<author>@chrisgleissner</author>
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from README.md to doc/developer.md, placing it after the Setup
section.
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</comment_thread>

</pr_reviews>

<pr_comments>
  <comment_new>
  <comment_id>3466481204</comment_id>
  <author>@chrisgleissner</author>
  @copilot Address all PR comments
  </comment_new>

</pr_comments>

</comments>

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The last **12** git commits in this branch are the changes you
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